

AI ASSISTANT CODING

ASSIGNMENT-11.5

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Lab 11 – Data Structures with AI: Implementing Fundamental Structures

Task Description #1 – Stack Implementation

Task: Use AI to generate a Stack class with push, pop, peek, and is_empty methods.

Sample Input Code:

```
class Stack:
```

```
    pass
```

Expected Output:

- A functional stack implementation with all required methods and docstrings.

```
"""Prompt:
Generate a Python class named Stack that implements the stack data
structure using a list. The class should include the following
methods:
push(item) - to add an element to the top of the stack
pop() - to remove and return the top element
peek() - to return the top element without removing it
is_empty() - to check whether the stack is empty
Include proper docstrings for the class and each method. Handle edge
cases such as popping from an empty stack."""
```

```
main.py  Run  Output
10 class Stack:
11     """
12     A class to represent a stack data structure.
13     Uses a Python list to store stack elements.
14     """
15     def __init__(self):
16         """Initialize an empty stack."""
17         self.items = []
18
19     def push(self, item):
20         """
21         Add an element to the top of the stack.
22         param [int] item: Element to be added to the stack
23         """
24         self.items.append(item)
25
26     def pop(self):
27         """
28         Remove and return the top element of the stack.
29         :return: Top element of the stack
30         :raises IndexError: If the stack is empty
31         """
32         if self.is_empty():
33             raise IndexError("Pop from empty stack")
34         return self.items.pop()
35
36     def peek(self):
37         """
38         Return the top element without removing it.
39         :return: Top element of the stack
40         :raises IndexError: If the stack is empty
41         """
42         if self.is_empty():
43             raise IndexError("Peek from empty stack")
44         return self.items[-1]
45
46     def is_empty(self):
47         """
48         Check whether the stack is empty.
49         :return: True if stack is empty, False otherwise
50         """
51         return len(self.items) == 0
52
53
54 if __name__ == "__main__":
55     stack = Stack()
56     stack.push(10)
57     stack.push(20)
58     stack.push(30)
59
60     print("Top element:", stack.peek()) # 30
61     print("Popped element:", stack.pop()) # 30
62     print("Is stack empty?", stack.is_empty()) # False
```

```
Top element: 30
Popped element: 30
Is stack empty? False
== Code Execution Successful ==
```

Task Description #2 – Queue Implementation

Task: Use AI to implement a Queue using Python lists.

Sample Input Code:

class Queue:

pass

Expected Output:

- FIFO-based queue class with enqueue, dequeue, peek, and size methods.

""" Prompt:
Generate a Python class named Queue that implements a FIFO (First-In-First-Out) queue using a Python list. The class should include the following methods:

enqueue(item) - to add an element to the rear of the queue
dequeue() - to remove and return the front element
peek() - to return the front element without removing it
size() - to return the number of elements in the queue """

```

main.py
11 class Queue:
12     """
13     A class to represent a FIFO (First-In-First-Out) queue.
14     Uses a Python list to store queue elements.
15     """
16     def __init__(self):
17         """Initialize an empty queue."""
18         self.items = []
19
20     def enqueue(self, item):
21         """
22         Add an element to the rear of the queue.
23         (params item: Element to be added to the queue)
24         """
25         self.items.append(item)
26
27     def dequeue(self):
28         """
29         Remove and return the front element of the queue.
30         (returns: front element of the queue)
31         raises IndexError: If the queue is empty
32         """
33         if self.is_empty():
34             raise IndexError("Dequeue from empty queue")
35         return self.items.pop(0)
36
37     def peek(self):
38         """
39         Return the front element without removing it.
40         (returns: front element of the queue)
41         raises IndexError: If the queue is empty
42         """
43
44     def size(self):
45         """
46         Return the number of elements in the queue.
47         (returns: number of elements in the queue)
48         """
49         return len(self.items)
50
51     def is_empty(self):
52         """
53         Check whether the queue is empty.
54         (returns: True if queue is empty, False otherwise)
55         """
56         return len(self.items) == 0
57
58 if __name__ == "__main__":
59     queue = Queue()
60
61     queue.enqueue(10)
62     queue.enqueue(20)
63     queue.enqueue(30)
64
65     print("Front element:", queue.peek()) # 10
66     print("Dequeued element:", queue.dequeue()) # 10
67     print("Queue size:", queue.size()) # 2

```

Output

```

Front element: 10
Dequeued element: 10
Queue size: 2
== Code Execution Successful ==

```

Task Description #3 – Linked List

Task: Use AI to generate a Singly Linked List with insert and display methods.

Sample Input Code:

class Node:

pass

class LinkedList:

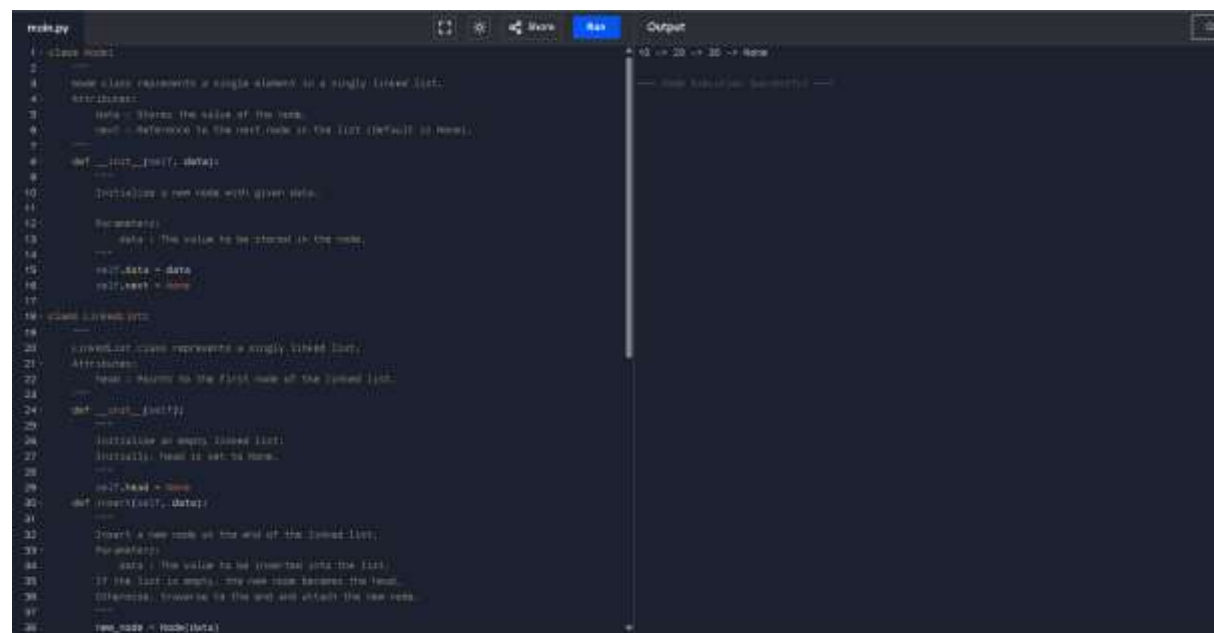
pass

Expected Output:

- A working linked list implementation with clear method documentation

```
""" Prompt:
Generate a Python implementation of a Singly Linked List.
Create a Node class to store data and a reference to the next node.
Create a LinkedList class with the following methods:

insert(data) - to insert a new node at the end of the list
display() - to display all elements of the linked list
Include clear docstrings for all classes and methods. """
```



```
1 class Node:
2     """
3     Node class represents a single element in a singly linked list.
4     Attributes:
5     data: Stores the value of the node.
6     next: Reference to the next node in the list (Default is None).
7     """
8     def __init__(self, data):
9         """
10        Initialize a new node with given data.
11        """
12        self.data = data
13        self.next = None
14
15 class LinkedList:
16     """
17     LinkedList class represents a singly linked list.
18     Attributes:
19     head: Points to the first node of the linked list.
20     """
21     def __init__(self):
22         """
23         Initialize an empty linked list.
24         Initially, head is set to None.
25         """
26         self.head = None
27
28     def insert(self, data):
29         """
30        Insert a new node at the end of the linked list.
31        """
32        new_node = Node(data)
33        if self.head is None:
34            self.head = new_node
35        else:
36            current = self.head
37            while current.next is not None:
38                current = current.next
39            current.next = new_node
```

```

14: data = 100 value to be inserted into the list.
15: If the list is empty, the new node becomes the head.
16: Otherwise, traverse to the end and attach the new node.
17:
18: new_node = Node(data)
19: # If list is empty, new node is the head
20: if self.head is None:
21:     self.head = new_node
22:     return
23: # Traverse to the last node
24: current = self.head
25: while current.next:
26:     current = current.next
27: # Link the last node to the new node
28: current.next = new_node
29:
30: def display(self):
31:     """Display all elements of the linked list.
32:     Traverse from head to the last node
33:     and prints each element followed by '\n'"""
34:     current = self.head
35:     if current is None:
36:         print("The list is empty.")
37:         return
38:     while current:
39:         print(current.data, end=" ")
40:         current = current.next
41:
42: if __name__ == '__main__':
43:     llist = LinkedList()
44:     llist.insert(10)
45:     llist.insert(20)
46:     llist.insert(30)
47:     llist.insert(40)
48:     llist.display()

```

Task Description #4 – Hash Table

Task: Use AI to implement a hash table with basic insert, search, and delete methods.

Sample Input Code:

class HashTable:

pass

Expected Output:

- Collision handling using chaining, with well-commented methods

```

1 """Task Description #4 - Hash Table
2 Task: Use AI to implement a hash table with basic insert, search, and delete
3 methods.
4 Sample Input Code:
5 class HashTable:
6     pass
7 Expected Output:
8 • Collision handling using chaining, with well-commented methods. """

```

```

1 class HashTable:
2     """
3     HashTable implementation using chaining for collision handling.
4     Attributes:
5     size: size of the hash table.
6     table: list of buckets (each bucket is a list)
7     """
8     def __init__(self, size=10):
9         """
10        Initialize the hash table with given size.
11        Parameters:
12        size (int): Number of buckets in the hash table.
13        """
14        self.size = size
15        self.table = [[] for _ in range(size)]
16
17    def search(self, key):
18        """
19        Returns method to compute hash index for a given key.
20        Parameters:
21        key: key to hash.
22        Returns:
23        int: index within table range.
24        """
25        return hash(key) % self.size
26
27    def insert(self, key, value):
28        """
29        Inserts a key-value pair into the hash table.
30        Handles collisions using chaining.
31        If the key already exists, updates its value.
32        """
33        index = self._hash(key)
34        bucket = self.table[index]
35        for pair in bucket:
36            if pair[0] == key:
37                pair[1] = value  # update existing key
38            return
39        bucket.append((key, value))  # add new key-value pair
40
41    def search(self, key):
42        """
43        Search for a key in the hash table.
44        Parameters:
45        key: key to search.
46        Returns:
47        value associated with key, or None if not found.
48        """
49        index = self._hash(key)
50        bucket = self.table[index]
51        for pair in bucket:
52            if pair[0] == key:
53                return pair[1]
54        return None  # key not found
55
56    def delete(self, key):
57        """
58        Deletes a key-value pair from the hash table.
59        Parameters:
60        key: key to delete.
61        Returns:
62        True if deleted successfully, False if key not found.
63        """
64        index = self._hash(key)
65        bucket = self.table[index]
66        for i, pair in enumerate(bucket):
67            if pair[0] == key:
68                del bucket[i]
69                return True
70        return False
71
72    if __name__ == '__main__':
73        ht = HashTable()
74        ht.insert('name', 'Softeq')
75        ht.insert('age', 25)
76        ht.insert('city', 'Azerbaijan')
77        print('Search name:', ht.search('name'))
78        print('Delete age:', ht.delete('age'))
79        print('Search age:', ht.search('age'))

```

Task Description #5 – Graph Representation

Task: Use AI to implement a graph using an adjacency list.

Sample Input Code:

class Graph:

pass

Expected Output:

- Graph with methods to add vertices, add edges, and display connections.

```
- """ Prompt:
2 Generate a Python class named Graph that represents a graph using an adjacency list.
3 The class should include the following methods:
4 add_vertex(vertex) - to add a new vertex to the graph
5 add_edge(vertex1, vertex2) - to add an edge between two vertices
6 display() - to display all vertices and their connected neighbors. """

class Graph:
    """A class representing a graph using an adjacency list.
    Attributes:
        vertices: A dictionary mapping vertex IDs to their neighbors.
        adj_list: A dictionary mapping vertex IDs to their neighbors.
    """
    def __init__(self):
        self.vertices = {}
        self.adj_list = {}

    def add_vertex(self, vertex):
        """Add a new vertex to the graph.
        Parameters:
            vertex: The vertex ID to be added.
        """
        self.vertices[vertex] = []
        self.adj_list[vertex] = []

    def add_edge(self, vertex1, vertex2):
        """Add an edge between two vertices.
        Parameters:
            vertex1: The first vertex ID.
            vertex2: The second vertex ID.
        """
        self.vertices[vertex1].append(vertex2)
        self.vertices[vertex2].append(vertex1)
        self.adj_list[vertex1].append(vertex2)
        self.adj_list[vertex2].append(vertex1)

    def display(self):
        """Display all vertices and their connected neighbors.
        Returns:
            A string representation of the graph.
        """
        output = ""
        for vertex in self.vertices:
            output += f"Vertex {vertex}: {self.vertices[vertex]}\n"
        return output
```

Task Description #6: Smart Hospital Management System – Data Structure Selection

A hospital wants to develop a Smart Hospital Management System that handles:

1. Patient Check-In System – Patients are registered and treated in order of arrival.
2. Emergency Case Handling – Critical patients must be treated first.
3. Medical Records Storage – Fast retrieval of patient details using ID.
4. Doctor Appointment Scheduling – Appointments sorted by time.
5. Hospital Room Navigation – Represent connections between wards and rooms.

Student Task:

- For each feature, select the most appropriate data structure from the list below:
 - o Stack
 - o Queue
 - o Priority Queue
 - o Linked List
 - o Binary Search Tree (BST)

- o Graph
- o Hash Table
- o Deque
- Justify your choice in 2–3 sentences per feature.
- Implement one selected feature as a working Python program with AI-assisted code generation.

Smart Hospital Management System – Data Structure Selection

1. Patient Check-In System

Selected Data Structure: Queue

Justification:

A Queue follows the First-In-First-Out (FIFO) principle. Patients are treated in the order they arrive, ensuring fairness. Therefore, a Queue is the most suitable structure for managing regular patient check-ins.

2. Emergency Case Handling

Selected Data Structure: Priority Queue

Justification:

Emergency patients must be treated based on severity rather than arrival time. A Priority Queue allows higher-priority (critical) patients to be treated first. This ensures immediate attention to life-threatening cases.

3. Medical Records Storage

Selected Data Structure: Hash Table

Justification:

Patient records need fast retrieval using a unique patient ID. A Hash Table provides average $O(1)$ time complexity for search, insert, and delete operations. This makes record management efficient and scalable.

4. Doctor Appointment Scheduling

Selected Data Structure: Binary Search Tree (BST)

Justification:

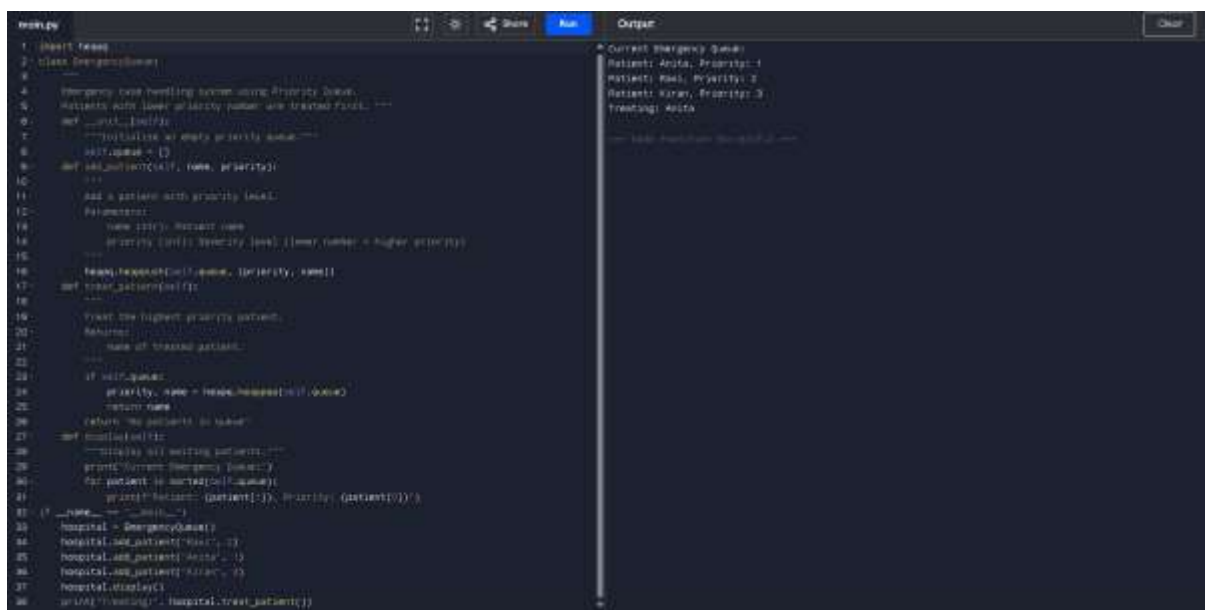
Appointments must be maintained in sorted order based on time. A Binary Search Tree automatically keeps elements sorted. This makes scheduling and time-based retrieval efficient.

5. Hospital Room Navigation

Selected Data Structure: Graph

Justification:

Hospital rooms and wards can be represented as nodes, and pathways as edges. A Graph effectively models connections between locations. It also supports pathfinding for navigation within the hospital.



```
1 # Import heapq
2 class EmergencyQueue:
3     """
4     Emergency queue handling system using Priority Queue.
5     Patients with lower priority number are treated first. ---
6     """
7     def __init__(self):
8         """Initialize an empty priority queue"""
9         self.queue = []
10        def add_patient(self, name, priority):
11            """
12            Add a patient with priority (lower).
13            Parameters:
14            name (str): Patient name
15            priority (int): Priority (lower number = higher priority)
16            """
17            heapq.heappush(self.queue, (priority, name))
18        def treat_patient(self):
19            """
20            Treat the highest priority patient.
21            Returns:
22            name of treated patient.
23            """
24            if self.queue:
25                priority, name = heapq.heappop(self.queue)
26                return name
27            return "No patients to treat"
28        def display(self):
29            """Display all waiting patients"""
30            print("Current Emergency Queue:")
31            for patient in sorted(self.queue):
32                print(f"Patient: {patient[1]}, Priority: {patient[0]}")
33        def __str__(self):
34            return str(self.queue)
35
36 # Example usage
37 hospital = EmergencyQueue()
38 hospital.add_patient("Ravi", 3)
39 hospital.add_patient("Anita", 1)
40 hospital.add_patient("Kiran", 2)
41 hospital.display()
42 print(f"Treating: {hospital.treat_patient()}")
```

Output:

```
Current Emergency Queue:
Patient: Anita, Priority: 1
Patient: Ravi, Priority: 3
Patient: Kiran, Priority: 2
Treating: Anita
```

Task Description #7: Smart City Traffic Control System

A city plans a Smart Traffic Management System that includes:

1. Traffic Signal Queue – Vehicles waiting at signals.
2. Emergency Vehicle Priority Handling – Ambulances and fire trucks prioritized.
3. Vehicle Registration Lookup – Instant access to vehicle details.
4. Road Network Mapping – Roads and intersections connected logically.

5. Parking Slot Availability – Track available and occupied slots.

Student Task

- For each feature, select the most appropriate data structure from the list below:
 - o Stack
 - o Queue
 - o Priority Queue
 - o Linked List
 - o Binary Search Tree (BST)
 - o Graph
 - o Hash Table
 - o Deque
- Justify your choice in 2–3 sentences per feature.
- Implement one selected feature as a working Python program with AI-assisted code generation.

Expected Output:

- A table mapping feature → chosen data structure → justification.
- A functional Python program implementing the chosen feature with comments and docstrings.

Smart City Traffic Control System

1. Traffic Signal Queue

Selected Data Structure: Queue

Justification:

Vehicles waiting at a traffic signal follow the First-In-First-Out (FIFO) principle. The vehicle that arrives first should move first when the signal turns green. Therefore, a Queue is the most appropriate structure for managing traffic signals fairly and efficiently.

2. Emergency Vehicle Priority Handling

Selected Data Structure: Priority Queue

Justification:

Emergency vehicles such as ambulances and fire trucks must be given immediate access regardless of arrival time. A Priority Queue allows higher-priority vehicles to be processed before regular vehicles. This ensures faster emergency response and improved public safety.

3. Vehicle Registration Lookup

Selected Data Structure: Hash Table

Justification:

Vehicle details must be accessed instantly using a registration number. A Hash Table provides average $O(1)$ time complexity for search operations. This makes it ideal for fast and efficient vehicle information retrieval.

4. Road Network Mapping

Selected Data Structure: Graph

Justification:

Roads and intersections can be represented as nodes and edges. A Graph efficiently models connections between locations and supports route-finding algorithms. This makes it suitable for navigation and traffic management systems.

5. Parking Slot Availability

Selected Data Structure: Binary Search Tree (BST)

Justification:

Parking slots can be stored in sorted order based on slot numbers. A Binary Search Tree maintains elements in sorted order and allows efficient insertion, deletion, and searching. This makes it suitable for tracking available and occupied parking slots.

```
main.py  Run  Output  Clear
1 class Stack:
2     """
3     Stack implementation using Python list.
4     Follows LIFO (Last-In-First-Out) principle. """
5     def __init__(self):
6         """
7         Initialize an empty stack.
8         """
9         self.items = []
10    def push(self, item):
11        """
12        Add an element to the top of the stack.
13        Parameters:
14        item (Element): Element to be added
15        """
16        self.items.append(item)
17    def pop(self):
18        """
19        Remove and return the top element of the stack.
20        Returns:
21        The top element of the stack.
22        Raises:
23        IndexError: If stack is empty. """
24        if self.is_empty():
25            raise IndexError("Pop from empty stack")
26        return self.items.pop()
27    def peek(self):
28        """
29        Return the top element without removing it.
30        Returns:
31        The top element of the stack.
32        Raises:
33        IndexError: If stack is empty. """
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3. Top-Selling Products Tracker – Products ranked by sales count.
4. Product Search Engine – Fast lookup of products using product ID.
5. Delivery Route Planning – Connect warehouses and delivery locations.

Student Task

- For each feature, select the most appropriate data structure from the list below:
 - o Stack
 - o Queue
 - o Priority Queue
 - o Linked List
 - o Binary Search Tree (BST)
 - o Graph
 - o Hash Table
 - o Deque
- Justify your choice in 2–3 sentences per feature.
- Implement one selected feature as a working Python program with AI-assisted code generation.

Expected Output:

- A table mapping feature → chosen data structure → justification.
- A functional Python program implementing the chosen feature with comments and docstrings.

Smart E-Commerce Platform – Data Structure Selection

1. Shopping Cart Management

Selected Data Structure: Linked List

Justification:

Products in a shopping cart are added and removed dynamically. A Linked List allows

efficient insertion and deletion without shifting elements. This makes it suitable for managing frequently changing cart items.

2. Order Processing System

Selected Data Structure: Queue

Justification:

Orders must be processed in the order they are placed. A Queue follows the First-In-First-Out (FIFO) principle, ensuring fairness. Therefore, it is ideal for handling customer orders.

3. Top-Selling Products Tracker

Selected Data Structure: Priority Queue

Justification:

Products need to be ranked based on sales count. A Priority Queue allows items with higher sales to be accessed first. This ensures efficient tracking of top-selling products.

4. Product Search Engine

Selected Data Structure: Hash Table

Justification:

Products are searched using unique product IDs. A Hash Table provides average $O(1)$ time complexity for search operations. This makes product lookup fast and efficient.

5. Delivery Route Planning

Selected Data Structure: Graph

Justification:

Warehouses and delivery locations can be represented as nodes, and roads as edges. A Graph efficiently models connections and supports shortest path algorithms. This makes it suitable for route optimization and logistics planning.

```
main.py  Run Output Clear
1 # Create a dictionary structure
2 # Dictionary: Product Name, Price, Stock, Year, Type
3 # First Value is Name, product name
4 products = {}
5 # Function to add a product
6 def add_product(product_id, name, price):
7     products[product_id] = {
8         'name': name,
9         'price': price
10    }
11    print('Product added successfully.')
12 # Function to search for a product
13 def search_product(product_id):
14     if product_id in products:
15         print('Product found:')
16         print('Product ID:', product_id)
17         print('Name:', products[product_id]['name'])
18         print('Price:', products[product_id]['price'])
19     else:
20         print('Product not found.')
21 # Function to display all products
22 def display_products():
23     if len(products) == 0:
24         print('No products available.')
25     else:
26         print('Available Products:')
27         for pid in products:
28             print(pid, '-', products[pid])
29 # Main Function
30 add_product(101, 'Laptop', 5000)
31 add_product(102, 'Mobile', 2000)
32 add_product(103, 'Headphones', 1000)
33 print()
34 display_products()
35 print('~Searching for Product 101~')
36 search_product(101)
37 print('~Searching for Product 200~')
38 search_product(101)
```

```
Product added successfully.
Product added successfully.
Product added successfully.

Available Products:
101 - {'name': 'Laptop', 'price': 5000}
102 - {'name': 'Mobile', 'price': 2000}
103 - {'name': 'Headphones', 'price': 1000}

Searching for Product 101

Product Found:
Product ID: 101
Name: Mobile
Price: 2000

Searching for Product 200

Product not found.

No product information is available.
```